EURECOM

Research Project Report

Multi-UE Mesh on OpenAirInterface: a 5G NR SL Architecture supporting Prototyping and Emulation

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Abstract

EURECOM's flagship 5G platform, OpenAirInterface (OAI), provides an RFSim (RF Simulator) mode to emulate propagation and communication conditions between multiple UEs. RFSim has recently been refactored to operate according to a publish/subscribe architecture, where the broker's role is to compute the propagation conditions and reception likelihood between a transmitter and multiple receivers. However, the current RFSim only provides basic ON/OFF strategies under static scenarios. The objective of this project is to enhance the RFSim broker by adding three new entities: propagation, mobility, and reception modules. The propagation module integrates various statistical propagation models to compute the attenuated energy at the receiver (RX), considering the transmitter (TX) energy, the distance between TX/RX entities, and potential background noise. The mobility module tracks the position, heading, and velocity of each entity, providing this data to the other modules. Finally, the reception module converts the Signal-to-Noise Ratio (SNR) to Bit Error Rate (BER) and reception likelihood according to the receiver's specifications. The system-level reception estimates are validated against OAI's internal polar decoder, achieving agreement within one percentage point across tested scenarios.

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List of Abbreviations

3GPP	Third Generation Partnership Project
API	Application Programming Interface
AVX-512	Advanced Vector Extensions 512-bit
AWGN	Additive White Gaussian Noise
BATMAN	Better Approach To Mobile Adhoc Networking
BER	Bit Error Rate
BLER	Block Error Rate
BW	Bandwidth
CFO	Carrier Frequency Offset
CLI	Command Line Interface
CP-OFDM	Cyclic Prefix Orthogonal Frequency-Division Multiplexing
CPU	Central Processing Unit
CQI	Channel Quality Indicator
CRC	Cyclic Redundancy Check
CSI-RS	Channel State Information Reference Signal
D2D	Device-to-Device
DMRS	Demodulation Reference Signal
DTX	Discontinuous Transmission
ETSI	European Telecommunications Standards Institute
FSPL	Free Space Path Loss
gNB	Next Generation NodeB (5G Base Station)
GPU	Graphics Processing Unit
HARQ	Hybrid Automatic Repeat reQuest
HPC	High Performance Computing
I/Q	In-phase and Quadrature (samples)

ICI	Inter-Carrier Interference
ICMP	Internet Control Message Protocol
InH	Indoor Hotspot
IoT	Internet of Things
IP	Internet Protocol
ITU-R	ITU Radiocommunication Sector
JSON	JavaScript Object Notation
LBT	Listen Before Talk
LDPC	Low-Density Parity-Check
LOS	Line Of Sight
LTE	Long Term Evolution
MAC	Medium Access Control
MCL	Maximum Coupling Loss
MCS	Modulation and Coding Scheme
MIMO	Multiple-Input Multiple-Output
NLOS	Non Line Of Sight
NLOSv	Non Line Of Sight (vehicle-blocked)
NR	New Radio
OAI	OpenAirInterface
OCUDU	Open Centralized Unit - Distributed Unit
OFDM	Orthogonal Frequency-Division Multiplexing
OGM	Originator Message (B.A.T.M.A.N.)
O-RAN	Open Radio Access Network
P2P	Peer-to-Peer / Pedestrian-to-Pedestrian
PC5	Proximity-based Communication Interface 5 (Sidelink interface)
PHY	Physical Layer
PLL	Phase-Locked Loop
PSBCH	Physical Sidelink Broadcast Channel
PSCCH	Physical Sidelink Control Channel
PSSCH	Physical Sidelink Shared Channel
pub/sub	Publish/Subscribe

QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
REQ/REP	Request/Reply
RF	Radio Frequency
RFSim	Radio Frequency Simulator
RWP	Random Waypoint
RX	Receiver
SF	Shadow Fading
SIMD	Single Instruction Multiple Data
SIMDe	SIMD Everywhere (portability library)
SINR	Signal-to-Interference-plus-Noise Ratio
SLSS	Sidelink Synchronization Signal
SNR	Signal-to-Noise Ratio
SUMO	Simulation of Urban MObility
SYNC-REF	Synchronization Reference
TBS	Transport Block Size
TCP	Transmission Control Protocol
TDMA	Time Division Multiple Access
TQ	Transmission Quality (B.A.T.M.A.N.)
TSDB	Time Series Database
TX	Transmitter
UDP	User Datagram Protocol
UE	User Equipment
URLLC	Ultra-Reliable Low-Latency Communication
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything
ZMQ	ZeroMQ (Message Queue library)

Chapter 1

Introduction

1.1 Motivation and Problem Formulation

EURECOM's flagship 5G platform OpenAirInterface (OAI) provides an RF Simulator (RFSim) mode that allows the execution of the full protocol stack without requiring a physical RF board, by sending time-domain I/Q samples over the network to emulate propagation and communication conditions between multiple UEs [8]. RFSim has recently been refactored to operate according to a pub/sub architecture, where the broker's role is to compute propagation conditions and reception likelihood between transmitters and receivers. Testing sidelink with multiple UEs requires several physical radio boards simultaneously, which are costly and scarce in a lab environment. RF simulation removes this constraint by emulating UE-to-UE propagation entirely in software.

The original broker implementation is minimal, functioning primarily as a ZMQ proxy without integrated channel intelligence, it performs pure message forwarding with no logic for mobility or channel modeling. The structural challenge lies in the fact that any existing fading or mobility logic is currently embedded within the radio's monolithic code, making it impossible to swap models dynamically or distribute simulation tasks across different machines.

In the traditional RFSim setup, a single entity acts as the RFSim server and all other entities connect as clients. This works well for infrastructure-based 5G scenarios where UEs communicate only with the gNB, but it becomes a problem for sidelink because UEs need to communicate directly with each other, which is not supported by the client-server model [9, 7, 1].

1.2 Proposed Solution

The objective of this project is to enhance the RFSim broker by adding three new entities: propagation, mobility, and reception modules. The three modules - propagation, mobility, and reception - are described in detail in Chapter 3.

Chapter 2

Platform Analysis

2.1 Description of OpenAirInterface (OAI)

OpenAirInterface (OAI) is an open-source software platform for 5G mobile telecommunications, reproducing the full system stack: UE software, base stations, core network, and communication protocols. The platform is maintained by the OpenAirInterface Software Alliance, a community of academic and industrial partners advancing research in radio access networks [2] [3].

2.2 Sidelink Protocol Stack

5G sidelink, in the form standardized by the Third Generation Partnership Project (3GPP) Release 16, is a feature of the 5G system that enables devices to communicate directly with one another without necessitating the need to route user data via the cellular infrastructure. Its use is growing in importance, especially in the commercial, public safety, and automotive (V2X) domains [6] [5] [4]. Figure 2.1 illustrates the full protocol stack deployed on each UE in this project. To establish a sidelink connection, synchronization information is broadcasted from a Synchronization Reference (SyncRef) UE and potentially decoded and accepted by other nearby UEs. By detecting the SLSS (SL Synchronization Signal), a nearby UE is able to extract critical pieces of information about the SyncRef UE. This synchronization procedure is required for establishing a 5G sidelink connection between multiple UEs [68]. At the physical layer, sidelink uses three dedicated channels. The Physical Sidelink Broadcast Channel (PSBCH) for synchronisation signals, the Physical Sidelink Control Channel (PSCCH) for scheduling assignments, and the Physical Sidelink Shared Channel (PSSCH) for user data. Resource allocation follows a TDMA scheme, a sidelink period spans 20 ms, within which each UE is assigned a non-overlapping time slot. The SYNC-REF UE anchors this schedule by pre-booking approximately half the available slots before any regular UE joins.

2.3 Description of the Existing RFSim

In the `sl-eurecom4` branch [9], a central broker replaces the traditional RFSim server. However, as shown in Figure 2.2, the broker is a passive ZMQ proxy with zero channel logic: all mobility and fading are embedded monolithically in `simulator.c` on the SYNC-REF side [10], controlled only via manual telnet commands at runtime. All UE pairs share the same static channel model instance, making true per-link independent fading impossible. In practice, the original broker supports only binary ON/OFF channel conditions. A link is either fully open or completely blocked, with no intermediate attenuation model. Furthermore, the architecture implicitly equates the SYNC-REF role with the gNB role, since the client-server topology assumes a single privileged server node, making true peer-to-peer D2D operation structurally impossible.

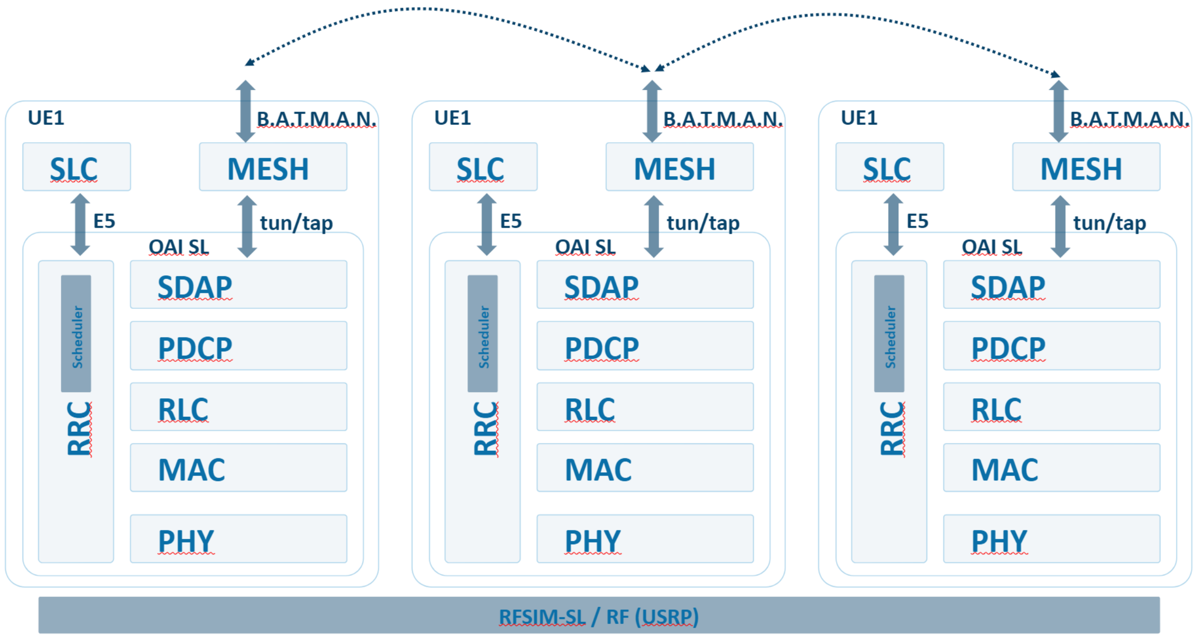


Figure 2.1: 5G NR Sidelink protocol stack for three UEs. Each UE runs the full OAI SL stack (PHY, MAC, RLC, PDCP, SDAP) with an RRC and MAC Scheduler. The tun/tap interface connects the OAI stack to the MESH layer, which runs B.A.T.M.A.N. for Layer-2 mesh routing across UEs. The shared transport layer at the bottom (RFSim-SL / RF USRP) is the component replaced or emulated by this project.

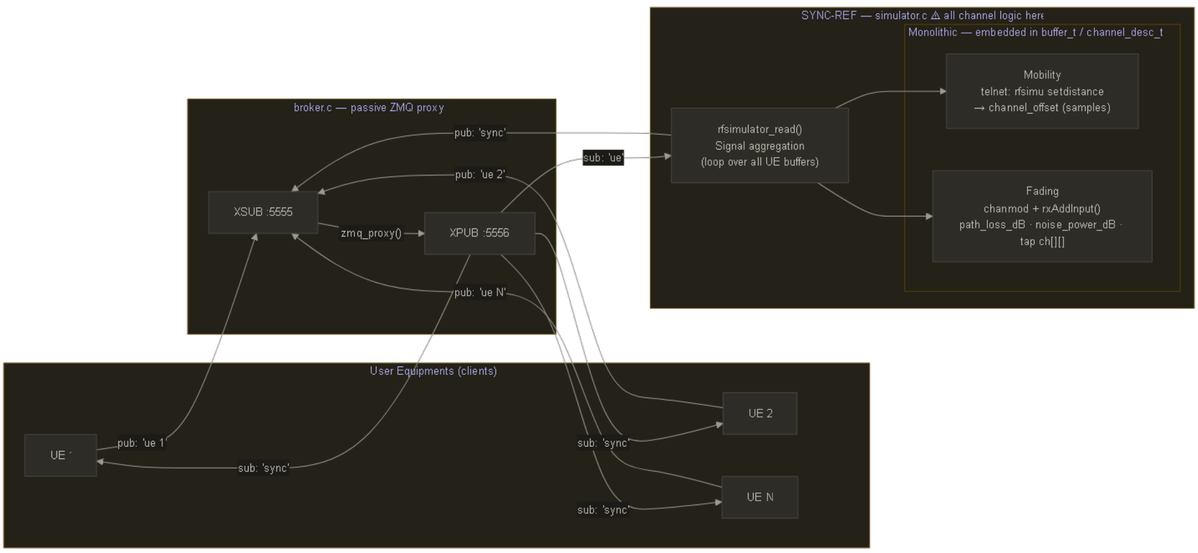


Figure 2.2: Architecture of the original RFSim (sl-eurecom4): broker.c is a passive ZMQ proxy with zero channel logic. All mobility (telnet `setdistance`) and fading (`chanmod`, `rxAddInput()`) are embedded monolithically in `simulator.c` on the SYNC-REF side.

2.4 Link-Level vs. System-Level Simulation

The original RFSim operates as a Link-Level Simulator, emulating the exact PHY behaviour of a 5G NR chipset by exchanging real I/Q samples - precise, but computationally expensive and limited to roughly 8–10 UEs in a sidelink lab setup. RFSim_SL instead try to target System-Level Simulation, channel intelligence is delegated to external modules that compute SNR, CQI, and BLER from validated 3GPP models, replacing the full PHY signal chain with a statistical abstraction that scales to hundreds of nodes. In a highway V2X scenario a single node may have 200–300 neighbours within range, making link-level simulation technically infeasible at that scale. Table 2.1 summarises the key trade-offs; the design of RFSim_SL is detailed in Chapter 3. It is important to underline that project continue using the link-level approach by level 2 of OAI, but this constitute a high level objective for the future.

Table 2.1: Link-Level vs. System-Level simulation: key trade-offs. In a highway V2X scenario, a single node may have 200–300 neighbours within range, making link-level simulation technically infeasible at that scale.

	Link-Level (RFSim)	**System-Level (future work)**
Max stable UEs	∼8–10 (lab)	200–300 (design target)
IQ processing	Full PHY chain	Statistical abstraction
Mobility	Manual telnet	Autonomous (RWP, SUMO)
Propagation model	Static path loss	3GPP TR 37.885/38.901
Configurability	Recompile required	Runtime API

This hierarchy of abstraction levels maps directly to scalability targets. At the full PHY simulation level (original RFSim), roughly 8–10 UEs are feasible in a controlled lab environment. At the statistical system level implemented in this work, the design target is on the order of 200 nodes, since expensive IQ processing is replaced by lightweight 3GPP formula evaluation. Future integration of geometric ray tracing (VRTSim, running on a separate GPU machine) would reduce this ceiling to roughly 20–25 nodes for scenarios requiring spatial precision, trading scalability for physical accuracy. The modular architecture of RFSim_SL is designed so that the propagation module can be replaced by VRTSim without changing the broker or any other component.

Chapter 3

New RF Simulator Design (RFSim_SL)

3.1 Architecture

The architecture must be a toolbox, you choose a mobility model, choose a fading model, configure it for an environment, and run. The mobility module and the propagation/fading module as external, independently callable entities, as well as on the refactoring of the broker to interact with them via a clean API. As illustrated in Figure 3.1, the system transitions from a monolithic structure to a decoupled environment where the mobility, propagation, and reception modules function as external, independently callable entities. Parameters of the modules can be edited at runtime using the CLI without restarting the server. All UEs, including SYNC-REF, are simultaneously transmitting broadcast packets. The broker should in principle route packets from every transmitter to every receiver.

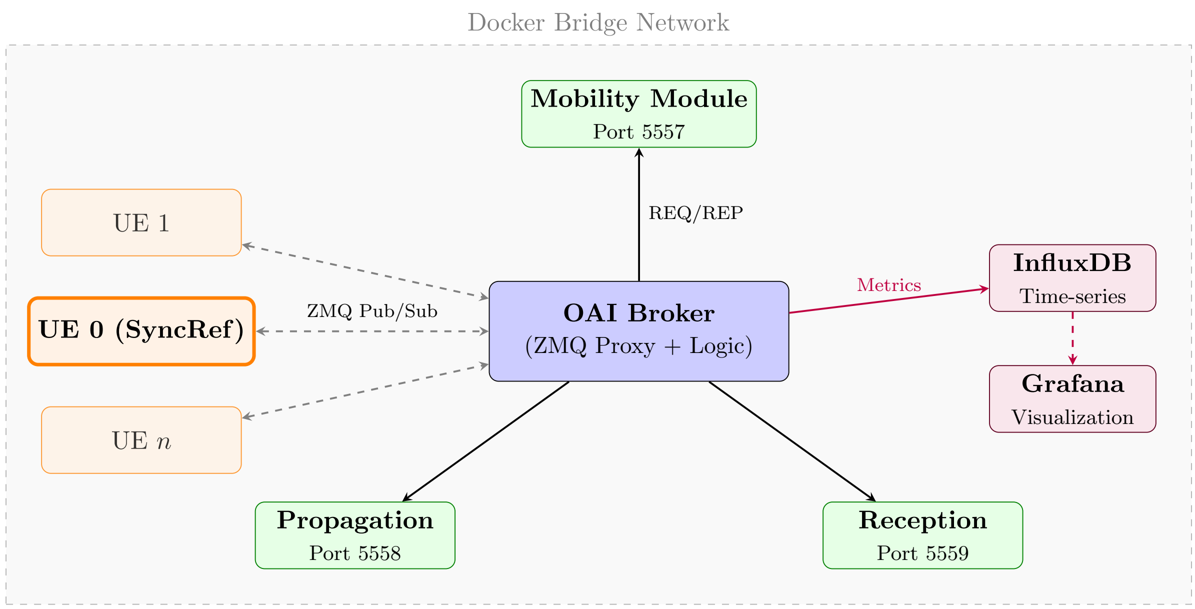


Figure 3.1: Modular Docker architecture for OAI simulations with Sidelink Mobility.

3.2 Module Descriptions

3.2.1 Mobility Module

The mobility module tracks position, heading, and velocity of each UE and implements Random Waypoint (RWP) mobility [11], replacing the manual telnet-based distance commands of the original RFSim.

The mobility module operates on a slower time scale than OAI, updating positions every 100,ms; the broker caches the result and extrapolates intermediate positions between refreshes.

The parameters that define a Random Waypoint scenario are the area side length (bounding box), the admissible speed range (minimum and maximum speed in $m \cdot s^{-1}$), and the maximum pause time at each waypoint. For every leg a UE draws a speed uniformly between the minimum and maximum values. Upon reaching a waypoint it pauses for a time chosen uniformly between zero and the configured maximum; the average stop time therefore equals half of that maximum.

The details of dynamic UE registration (ADD_UE) are described in Section 3.5.

The maximum pool size is 8 (matching MAX_SL_UE_CONNECTIONS [18]), but can be reconfigured at startup via –max-ues.

The mobility module runs as an independent process or Docker container, communicating with the broker via ZMQ REQ/REP with JSON payloads.

Runtime model switching is supported via the `SET_MODEL` API, allowing transitions between Random Waypoint and Static (frozen) mode without restarting any process.

Unlike what one might expect, `pymobility` [11] does not perform time-based interpolation internally; the broker therefore maintains its own linear extrapolation between 100,ms cache refreshes [21].

3.2.2 Propagation Module

It implements five 3GPP-compliant path loss models, each with probabilistic LOS/NLOS selection (Table 3.1). It runs as an independent process (or Docker container). The broker sends a JSON request and this module replies with path_loss_dB and noise_floor_dB for a specific TX→RX link.

A critical requirement is the ability to configure individual links between pairs of UEs. This means being able to set an arbitrary attenuation value for a specific UE-to-UE link independently of physical distance, as well as completely block a link, for example in the presence of a wall between two UEs.

The current state of the art in 3GPP and academic publications for sidelink channel modeling covers only two environments: highway and urban [34]. No other scenarios (e.g., suburban, industrial, indoor) have standardized sidelink propagation formulas. The limitation of being restricted to highway and urban has been documented explicitly in the fading module documentation. This is a known constraint inherited from the state of the art, not an implementation shortcoming.

The fading module is stateless with respect to UE positions: it receives distance and speed per query and returns path loss. Table 3.1 summarises the path loss coefficients, model parameters and LOS probability types for all five models; the non-obvious modeling choices for each are described below.

On every `COMPUTE_PATHLOSS` call, the module draws$u \sim \mathcal{U}(0,1)$ and selects the channel condition by comparing u against cumulative probability thresholds: $u < P(\text{LOS})$ gives LOS, $u < P(\text{LOS}) + P(\text{NLOSv})$ gives NLOSv (vehicle blockage), otherwise NLOS (building blockage). Both $P(\text{LOS})$ and $P(\text{NLOSv})$ decrease with distance, so short-range links are almost always LOS and long-range links almost always NLOS. Because the draw is independent at every 100 ms query, the channel condition can flip between LOS and NLOS at every broker update, faster than the physical coherence time of real environments. Long-term statistics are unbiased;

Table 3.1: Propagation models and path loss coefficients implemented in RFSim_SL [30, 31, 35, 37, 38].

Model	**Standard**	**Range**	**PL$_{\mathbf{LOS}}$ (dB)**	**PL$_{\mathbf{NLOS}}$ (dB)**	σ_{SF}	**LOS/NLOS**
Free Space	ITU-R P.525-3	No limit	$20\log d + 20\log f - 27.6$	–	0	LOS only
Urban V2V	3GPP TR 37.885	≤1000 m	$38.77 + 16.7\log d + 18.2\log f_c$	$36.85 + 30\log d + 18.9\log f_c$	3 / 4 dB	Exp.
Highway V2V	3GPP TR 37.885	≤1000 m	$32.4 + 20\log d + 20\log f_c$	$36.85 + 30\log d + 18.9\log f_c$	3 / 4 dB	Piecewise
Indoor (InH)	3GPP TR 38.901	1–150 m	$32.4 + 17.3\log d + 20\log f_c$	max(LOS, $17.3 + 38.3\log d + 24.9\log f_c$)	3 / 8 dB	Piecewise exp.
Rural	ECC-68 / Car2Car	3-seg.	d_{T0}=128 m, n_0=2.8, d_{T1}=512 m, n_1=3.3	LOS only	–	LOS only

instantaneous path loss carries higher variance (see Table 5.1). Hard blockage is enforced separately via `SET_LINK_BLOCK`.

The following paragraphs detail the non-obvious modeling choices for each model; full formula derivations and coefficient tables are available in the cited standards.

Free Space follows ITU-R P.525-3 [30] and serves as a theoretical baseline with no LOS/NLOS selection; it is used to isolate the effect of distance without environmental effects.

Urban V2V (3GPP TR 37.885 [31]) defines three channel states - LOS, NLOSv (vehicle blocker), and NLOS (building blocker) - selected probabilistically per query according to a density-dependent exponential function [33, 58]; the NLOSv additional loss A_{NLOSv} depends on the relative antenna heights of TX, RX, and the blocking vehicle via three cases [55], and NLOS is enforced via `SET_LINK_BLOCK` [32].

Highway V2V (3GPP TR 37.885 [31]) uses the same path loss expressions as Urban for LOS/NLOS but a piecewise quadratic LOS probability that decays gradually beyond 475 m [33, 58]; the same three-case A_{NLOSv} model applies [55, 62, 61].

Indoor (InH) (3GPP TR 38.901 [35]) applies a piecewise exponential LOS probability valid from 1 to 150 m and takes $PL_{\mathrm{NLOS}} = \max(PL_{\mathrm{LOS}}, PL'_{\mathrm{NLOS}})$ to enforce the physical constraint that NLOS loss cannot be lower than LOS [36].

Rural adopts the ECC Report 68 three-segment formula as parameterised by the Car2Car Consortium for vehicle-height antennas [37, 38], independently validated in OMNeT++ vehicular studies [63]; no 3GPP-standardised rural sidelink model exists, and NLOS conditions are handled via `SET_LINK_BLOCK` [32].

NS3 Cross-Validation

NS3 [64, 59] provides an independent C++ implementation of TR 37.885 against which the Python module was validated. As confirmed by the NR V2X tutorial presented at WNS3 2022 [50] (slide 6, channel model comparison table), the NS3 NR V2X module currently supports only two channel models that are genuinely derived from UE-to-UE measurements and are therefore applicable to true 5G NR sidelink (PC5) links: **V2V Urban** and **V2V Highway**, both from 3GPP TR 37.885 [64].

Shared Specification and Formula-Level Verification Inspecting the ns-3 C++ source [55, 53, 54] confirms an exact coefficient-level match for all deterministic path loss formulas and for the A_{NLOSv} log-normal parameterization [59].[1]

Intentional Discrepancies in Low-Density Scenarios The four mismatches in Table 3.2 are confined to the LOW density configurations: this module uses the TR 37.885 baseline formula [33] for low density, while NS3 applies Boban 2016 coefficients universally [58, 64, 56, 57].

Component	**OAI Python**	**NS-3 C++**	**Status**
All deterministic PL formulas (Urban/Highway LOS, NLOS, NLOSv) and shadow fading σ_{SF} [55, 53, 54, 62, 59]	-	exact coefficient match	MATCH
Urban P(LOS) LOW	(1.05, 0.0114)	(0.8548, 0.0064)	MISMATCH
Urban P(NLOSv) LOW	$1 - P$(LOS), two-state	lognorm kernel (0.0396, 5.2718, 3.4827)	MISMATCH
Highway P(LOS) LOW	Giordani piecewise [33]	$1.5 \times 10^{-6}d^2 - 0.0015d + 1$	MISMATCH
Highway P(NLOSv) LOW	$1 - P$(LOS), two-state	$-2.9 \times 10^{-7}d^2 + 5.9 \times 10^{-4}d + 0.0017$	MISMATCH

Table 3.2: NS-3 validation summary [64, 58, 56, 57]. All four mismatches are confined to the LOW density scenario: this module uses the TR 37.885 baseline [33] while NS-3 applies Boban 2016 coefficients universally.

3.2.3 Reception Module

The broker operates correctly without this module. IQ scaling, Doppler rotation, and per-link forwarding all function independently of any reception estimate. The module exists purely for observability. It makes visible the channel quality metrics using a specific pipeline computes BLER but does not influence the decision of what happens in layer 2 of OAI.

Converts the SNR (Signal-to-Noise Ratio) to Bit Error Rates (BER) and reception likelihood according to the receiver specification. Reception module is only monitoring, RFSim already handles reception internally [47]. Packet dropping remains inside OAI's PHY layer - specifically the PSBCH polar decoder CRC check in `nr_psbch_rx.c` [25, 47] - and is never duplicated in the broker.

The reception module uses as primary BLER source the UWICORE NR V2X SL dataset [43], containing 6,188 BLER vs. SNR curves following 3GPP Release 16 specifications [42, 31, 44]. Henderson et al. [65] demonstrate that BLER curves are frequency-independent once path loss is subtracted, validating the dataset at the OAI carrier of 2585,MHz. Most "sidelink" SNR–BLER curves in literature refer to LTE, not 5G NR [26]. The two are not interchangeable, and LTE curves can significantly mislead NR sidelink simulations; NR sidelink hardware is still mostly experimental.

[1] For point-by-point numerical comparison, shadow fading can be disabled via `--no-shadow`; the deterministic coefficients then match ns-3 exactly [51, 52].

The complete pipeline converting SNR to *rx_prob* through six steps (Doppler ceiling check, SNR penalty, CQI mapping, MCS mapping, BLER lookup, and reception probability) is illustrated in Figure 3.2.

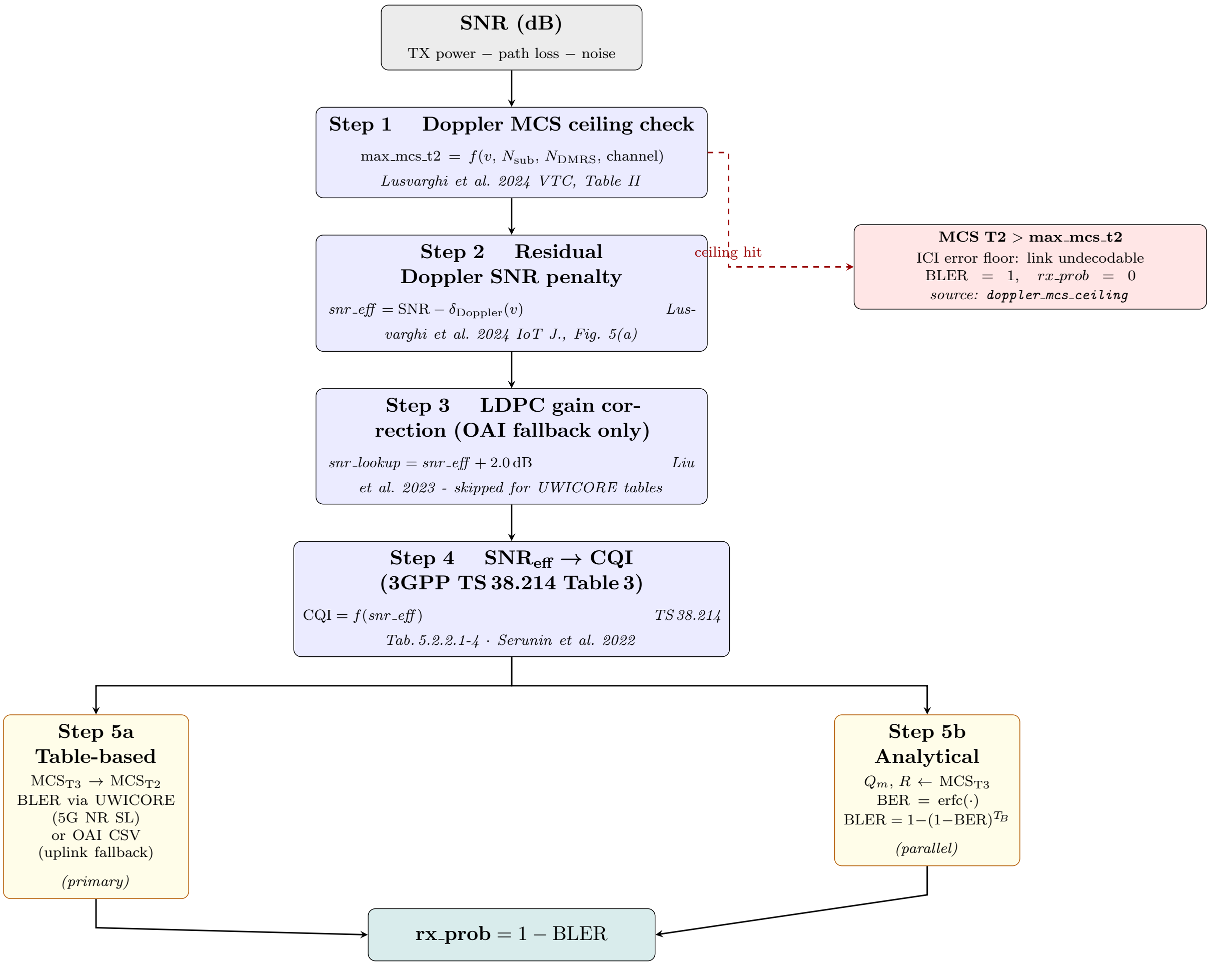


Figure 3.2: Reception probability pipeline: from raw SNR to *rx_prob* (v3). Step 1 checks whether the selected MCS exceeds the Doppler ICI ceiling (Paper [45]); if so, $rx\_prob = 0$ is set immediately without further computation. Otherwise, the residual Doppler SNR penalty (Step 2, Paper A [43]) is applied before the CQI lookup. Steps 5a (table-based, primary) and 5b (analytical, parallel) converge to the same output. The LDPC correction in Step 3 is applied only when the OAI uplink fallback tables are used.

Empirical Approach

Doppler MCS Ceiling At high relative speeds, Doppler shift on CP-OFDM produces an ICI error floor that no increase in TX power can overcome.

The reception module implements a *Doppler MCS ceiling* derived from [45], Table II. For each combination of channel model, v_{rel}, N_{sub}, and N_{DMRS}, the table identifies the maximum MCS T2 index achieving BLER ≤ 0.01; above this ceiling, BLER $= 1.0$ and rx_prob $= 0$ are set directly.

For example, at 280 km/h with $N_{\text{sub}} = 2$, $N_{\text{DMRS}} = 2$ (Highway LOS), the ceiling is MCS T2 $= 5$; any CQI ≥ 10 yields rx_prob $= 0$ regardless of SNR.

Residual Doppler SNR Penalty For MCS indices *below* the ceiling - i.e. when the link is physically decodable - Doppler still degrades performance by increasing the effective noise floor.

This residual effect is modelled as a speed-dependent SNR penalty $\delta_{\text{Doppler}}(v)$ applied before all subsequent pipeline steps [46]:

$$\text{SNR}_{\text{eff}} = \text{SNR} - \delta_{\text{Doppler}}(v)$$

The penalty values are read from Paper A (Lusvarghi et al. 2024, IEEE IoT J.) [43], Fig. 5(a), which reports the SNR required to reach BLER = 0.01 for QPSK-308, Highway LOS, $N_{\text{DMRS}} = 2$ at four relative speeds: 0, 1.8, 3.8, and 7.3 dB at $v_{\text{rel}} \in \{0, 70, 140, 280\}$ km/h respectively; intermediate values are linearly interpolated.

SNR To CQI The mapping follows 3GPP TS 38.214 CQI Table 3 [42], adopted by OAI for NR sidelink scheduling [39, 40] and confirmed for sidelink with one MIMO layer by Serunin et al. [41]. SNR thresholds are derived from OAI's `nr_csi_rs_cqi_estimation()` [40] with a +2 index offset; the full mapping is in Table 3.3. For URLLC vertical, a CQI Table 3 for lower BLER (Block Error Rate), is introduced to support URLLC service enabling the highly reliable link adaption for the sidelink requirements [48].

CQI to MCS The OAI sidelink scheduler uses MCS Table 3 [42, 39], so CQI is first mapped to MCS T3. Since the UWICORE dataset [43] is indexed by MCS Table 2, a second translation matches (Q_m, R) pairs between the two tables - a bookkeeping step introducing no physical approximation. CQI 0–3 carry no MCS assignment (BLER ≈ 1.0 below -20 dB); the complete mapping is in Table 3.3.

Table 3.3: Combined SNR-to-CQI and CQI-to-MCS mapping (3GPP TS 38.214 Tables 5.2.2.1-4, 5.1.3.1-3, 5.1.3.1-2 [42]). CQI 0–3 carry no MCS assignment; MCS T2 indexes the UWICORE BLER dataset [43].

CQI	SNR (dB)	Mod.	CR/1024	Eff.	MCS T3	MCS T2
0	< -20	-	-	-	-	-
1	$[-20, -6)$	QPSK	30	0.0586	-	-
2	$[-6, -3)$	QPSK	50	0.0977	-	-
3	$[-3, 0]$	QPSK	78	0.1523	-	-
4	$(0, 2]$	QPSK	120	0.2344	6	0
5	$(2, 3]$	QPSK	193	0.3770	8	1
6	$(3, 5]$	QPSK	308	0.6016	10	2
7	$(5, 7]$	QPSK	449	0.8770	12	3
8	$(7, 9]$	QPSK	602	1.1758	14	4
9	$(9, 10]$	16QAM	378	1.4766	16	5
10	$(10, 12]$	16QAM	490	1.9141	18	7
11	$(12, 15]$	16QAM	616	2.4063	20	9
12	$(15, 16]$	64QAM	466	2.7305	22	11
13	$(16, 18]$	64QAM	567	3.3223	24	13
14	$(18, 19]$	64QAM	666	3.9023	26	15
15	> 19	64QAM	772	4.5234	28	17

How To Obtain BLER and Reception Probability The UWICORE loader queries the curve indexed by (channel, v_{rel}, N_{sub}, N_{DMRS}, MCS T2) and interpolates linearly to obtain

BLER at the effective SNR. The reception probability is [66] [65]: [2]

$$P_{rx} = 1 - \text{BLER}(\text{channel}, v_{\text{rel}}, N_{\text{sub}}, N_{\text{DMRS}}, \text{MCS\_T2}, \text{SNR}_{\text{eff}})$$

Analytical Mathematical Approach

The table-based approach using link-level simulation datasets (UWICORE [43]) is the industry standard for PHY abstraction in system-level simulators [66, 65]: it accounts for LDPC coding, channel estimation errors, and hardware-specific effects that closed-form formulas cannot capture. As a parallel diagnostic reference, the module also computes BLER from closed-form BER expressions for Gray-coded QAM in AWGN [42]:

$$\text{QPSK:} \quad \text{BER} = \tfrac{1}{2}\,\text{erfc}\Big(\sqrt{SNR \cdot R}\Big) \tag{3.1}$$

$$\text{16-QAM:} \quad \text{BER} = \tfrac{3}{4}\,\text{erfc}\left(\sqrt{\tfrac{SNR \cdot R}{5}}\right) \tag{3.2}$$

$$\text{64-QAM:} \quad \text{BER} = \tfrac{7}{12}\,\text{erfc}\left(\sqrt{\tfrac{SNR \cdot R}{21}}\right) \tag{3.3}$$

where R is the normalised code rate ($R = \text{code rate} \times 1024\ /\ 1024$) taken from MCS Table 3 (3GPP TS 38.214 Table 5.1.3.1-3 [42]), and SNR is the linear signal-to-noise ratio. From the BER estimate, the block-level quantities follow as:

$$\text{BLER} = 1 - (1 - \text{BER})^{T_{BS}}, \qquad rx\_prob = 1 - \text{BLER}$$

where T_{BS} is the Transport Block Size in bits.

At 64-QAM, the `erfc` argument causes floating-point underflow (BER = 0, rx_prob = 1 regardless of code rate); the table-based path [43] is therefore always used as the primary method.

3.2.4 Broker

It remains simple and lightweight. Its role is only message routing and coordination. The broker is the central knowledge point, it knows which UEs are connected and what the channel conditions are between each pair.

Broker contains no channel logic. All intelligence is delegated to external modules via simple API calls. When no external modules are running, the broker automatically falls back to passthrough mode (original behaviour).

The broker computes per-link channel parameters independently for each pair and publishes on dedicated `sync_N` and `p2p_<rx>_<tx>` topics.

The workflow is the follow:

1. The broker caches the positions (including speed and heading) of all UEs by querying the mobility module every 100 ms. Distance and relative radial speed are then computed on-the-fly from these cached positions for each IQ block.

2. It queries the propagation module for the path loss and the Doppler shift, given the current distance and radial speed.

3. The broker applies an OFDM-aware, per-symbol Doppler phase rotation to the IQ samples (`apply_doppler_per_symbol`), using the Doppler shift returned by the propagation module. The phase is held constant within each OFDM symbol and advances between symbols

[2] `get_bler()` returns the last non-zero measured value for SNR above the table maximum, giving $rx\_prob \geq 0.9999$, a conservative lower bound (measurement floor $\approx 2.5 \times 10^{-5}$).

in proportion to f_d, modelling the intra-block ICI caused by relative motion. The phase resets to zero at each 2 ms block boundary, consistent with the OAI RFSim assumption of ideal carrier frequency tracking between blocks (i.e., no residual CFO compensation on the sidelink path) [67]. This approach correctly degrades reception quality as a function of relative speed:

- at $f_d = 0\,\text{Hz}$, the rotation is the identity;
- at $f_d = 100\,\text{Hz}$, the phase ramps by approximately 76° per block, reducing PSBCH success from approximately 73% to approximately 22%;
- at $f_d = 166\,\text{Hz}$, the original per-symbol implementation produced a 120° ramp that destroyed PSCCH decoding entirely.

4. It then applies the differential path-loss scaling (MCL convention) [20] to the already rotated IQ samples.
5. The processed IQ samples are forwarded to the receiver.
6. Every `LOG_INTERVAL` forwarded blocks, the broker queries the reception module with the computed SNR, receiving CQI, BER, and reception probability estimates in return.

The broker log format and all per-link fields are documented in the repository API specification [13]; the full processing flow is shown in Figure 3.3a.

Differential Path-Loss Scaling and I/Q Quantization

Applying absolute 3GPP path loss directly to OAI's 16-bit integer I/Q samples (normalized amplitude ≈ 256) causes numerical truncation to zero even at short range.

The broker adopts the Minimum Coupling Loss (MCL) convention [49, 20], exposed as a runtime CLI parameter (`--mcl`):

$$PL_{\text{effective}} = \max(0,\, PL_{\text{actual}} - \text{MCL})$$

This effective attenuation in decibels is then converted into a linear scalar factor and applied to the I/Q samples:

$$s_{out}[n] = s_{in}[n] \times 10^{\frac{-PL_{\text{effective}}}{20}}$$

Empirical calibration during this project demonstrated that setting the MCL to 80 dB optimally aligns the I/Q sample amplitude with the theoretical SNR, minimizing the divergence between the link-level OAI decoder and the system-level UWICORE tables.

OFDM-Aware Doppler Phase Rotation

Implementation The phase resets at each 2 ms block boundary, consistent with OAI RFSim's assumption of no residual CFO on the sidelink path [67]:

$$\varphi_k = \varphi_{k-1} + 2\pi f_d \cdot T_{\text{sym},k}$$
$$s_{\text{out}}[n] = s_{\text{in}}[n] \cdot e^{j\varphi_k} \quad \forall n \in \text{symbol } k$$

Validated Behaviour Measured on the EURECOM JupyterHub cluster (highway model, area = 200 m, 2 UEs): in passthrough mode PSBCH success is ≈100%; with channel-aware mode but Doppler disabled it drops to ≈73%, recovering to ≈81% at $f_d \approx 1\,\text{Hz}$. At $f_d = 100\,\text{Hz}$ success falls to ≈22%; the earlier per-symbol phase accumulation at $f_d = 166\,\text{Hz}$ caused complete failure (Batman = ×).

The degradation at 100 Hz is physically correct: the 76° intra-block phase ramp produces ICI that makes initial PSBCH acquisition harder, while subsequent PSCCH and PSSCH decoding recover because DM-RS-based channel estimation compensates the per-symbol phase offset within each slot.

Design Choice: Phase Reset per Block An earlier implementation accumulated phase across blocks, causing complete failure even at 1.7 Hz: OAI RFSim sidelink has no pilot-based CFO correction on the receive path, so the accumulated offset exceeded the PSBCH correlator coherence within a few hundred milliseconds. The per-block reset is equivalent to assuming the receiver removes the bulk carrier offset before each block, the standard PLL assumption.

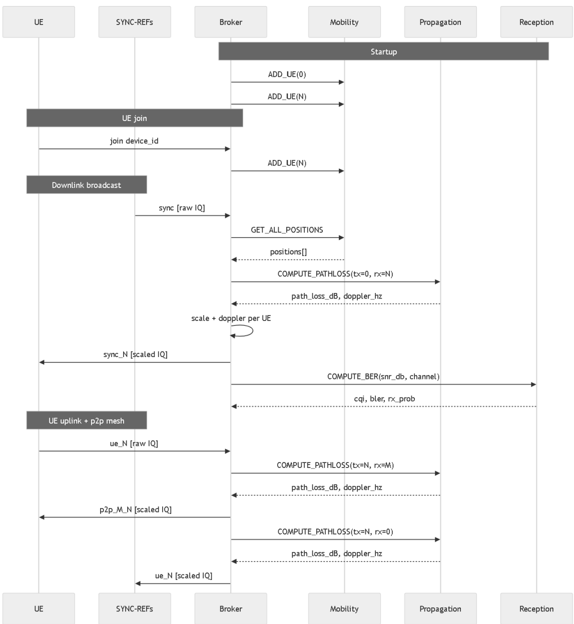
Analytical SNR Estimation

The broker computes SNR analytically rather than measuring received signal power [24]::

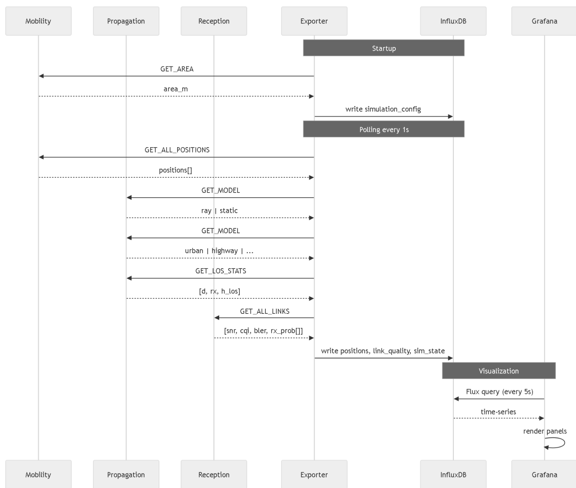
$$SNR_{dB} = P_{TX} - PL_{dB} - (N_0 + 10\log_{10}(BW))$$

where $P_{\mathrm{TX}} = 23\,\mathrm{dBm}$, $N_0 = -174\,\mathrm{dBm/Hz}$, and BW is the operational bandwidth.

Figures 3.3a and 3.3b illustrate the two communication patterns between the broker and the external modules.



(a) IQ block processing pipeline. The broker refreshes position cache every 100 ms, queries Propagation for path loss and Doppler, applies IQ scaling per link, and queries Reception every 500 blocks. Full-mesh p2p_M_N routing is also shown.



(b) Dashboard monitoring pipeline. The Exporter polls all modules every second, writes `ue_positions`, `link_quality`, `channel_conditions`, and `sim_state` to InfluxDB; Grafana reads via Flux queries every 5 s.

Figure 3.3: Sequence diagrams of the two main communication patterns: IQ block processing (left) and dashboard monitoring (right).

3.2.5 Grafana and InfluxDB Monitoring Stack

OAI is generally poor at exposing monitoring information on the sidelink side; the reception module fills this gap by computing SNR, BER, and CQI values that are otherwise invisible inside OAI's internal processing pipeline. Grafana [14, 15] was chosen as the visualization layer for its native support for time-series data sources and real-time dashboard rendering, backed by InfluxDB [16, 17] as the time-series database for its high write throughput and native support for timestamped metrics at sub-second granularity. The dashboard implements four panels: a 2D map with real-time UE positions, directional links, and LOS/NLOS state per pair; per-link SNR, CQI, BER, and BLER time-series with UE selection via dropdown; PSCCH statistics comparing the layer-2 reception lock ratio against the system-level model prediction; and distance and LOS fraction panels for correlation between physical proximity and channel metrics.

The monitoring layer intentionally avoids instrumenting OAI's internal code. Adding Grafana hooks directly inside OAI's core PHY procedures would require modifying files that are actively maintained upstream, creating conflicts at every merge and making the branch harder to maintain. Instead, the broker computes SNR analytically from the path loss returned by the propagation module and forwards it to the external reception server; Grafana then reads from an InfluxDB instance populated entirely outside the OAI process. The result is a monitoring stack that is fully decoupled from OAI's internals. It can be updated, replaced, or removed without touching a single line of PHY or MAC code.

3.3 Inter-Module Communication and API Specification

Table 3.4 summarises the command set defined in the technical specification [13].

Table 3.4: Modular RF Simulator API Specification[13].

Module	Command	Description
Mobility	`GET_ALL_POSITIONS`	Returns x, y coordinates, speed, and heading for all active UEs.
(Port 5557)	`SET_POSITION`	Manually teleports a UE, overriding internal movement logic.
	`ADD_UE`	Dynamically registers a new UE slot in the simulation pool.
Propagation	`COMPUTE_PATHLOSS`	Calculates attenuation (PL_{dB}) and Doppler shift for a TX→RX link.
(Port 5558)	`SET_MODEL`	Changes environmental logic (e.g., Urban, Highway, Rural, Indoor).
	`SET_LINK_BLOCK`	Forcibly blocks a directional link to simulate total obstruction.
Reception	`COMPUTE_BER`	Maps SNR to CQI and provides a reception probability estimate.
(Port 5559)	`GET_CQI_TABLE`	Returns the internal 3GPP-compliant SNR-to-CQI conversion table.

3.4 Docker Containerization

The original Linux namespace setup requires multiple terminals and manual IP/MAC assignment per UE. Docker replaces this with the following architecture:

- **One container per node:** broker, SYNC-REF, and each UE run in isolated containers on a shared Docker bridge network; Grafana and InfluxDB add two further containers.
- **Scalable:** adding a UE means launching one more container - no manual namespace or IP configuration.
- **Safe:** if the broker overloads, killing and restarting its container leaves all UE containers unaffected.
- **Portable:** the compiled OAI image can be pushed to Docker Hub and pulled on any machine without recompilation [12].

- **Automated launch:** the full stack (broker, SYNC-REF, N UE containers, monitoring) starts and tears down with a single command.

3.5 Dynamic UE Registration

UEs do not all need to be started simultaneously. To ensure a scalable and automated simulation environment, the association between OAI radio entities and the mobility module is fully dynamic. When a new UE connects to the broker via a `join` message, the broker automatically invokes the mobility module API to initialize a new movement trajectory for that specific entity.

The mobility server starts empty; each `ADD_UE` call creates a new entity at a random position within the simulation area and assigns it a Random Waypoint trajectory with randomised speed and heading. Each registration activates one slot in the pre-allocated pool (up to 8 UEs, matching `MAX_SL_UE_CONNECTIONS` [18]).

Chapter 4

Demo and Experimentation

4.1 Results from Multi-UE Tests

All experiments were conducted on the EURECOM JupyterHub cluster (Intel Xeon Gold, 8 cores, 8 GB RAM) running the channel-aware broker with the Urban V2V propagation model and Random Waypoint mobility. Tests ranged from the basic SYNC-REF + 2 UE configuration up to 15 simultaneously connected UEs. In all configurations, PSBCH synchronisation was confirmed (`PSBCH RX ok` incrementing) and Batman-adv Layer-2 routing converged within 3–5 minutes. Scalability results are reported in Section 4.4.2; the Grafana dashboard output is shown in Section 4.2. In all tested configurations, the broker operates in full-mesh mode by default. Every UE, including SYNC-REF, transmits to and receives from all other connected UEs simultaneously. At minimum, PSBCH synchronisation broadcasts are exchanged across all pairs, confirming basic sidelink connectivity in the mesh.

4.2 Grafana Dashboard Demonstration

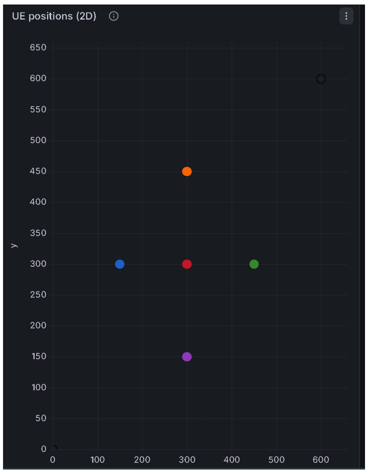
The dashboard is organised around link pairs (UEi → UEj); a dropdown selects the reference UE for all time-series panels. The topology view (Fig. 4.2) shows real-time UE positions and active links; per-link metrics (SNR, CQI, BER, *rx_prob*) are shown in Figs. 4.3 and 4.1, and the system-level vs. PHY-layer validation is in Fig. 4.4.

(a) Propagation model, mobility model. (b) Vehicle density, LOS fraction. (c) BER over time. (d) *rx_prob* over time.

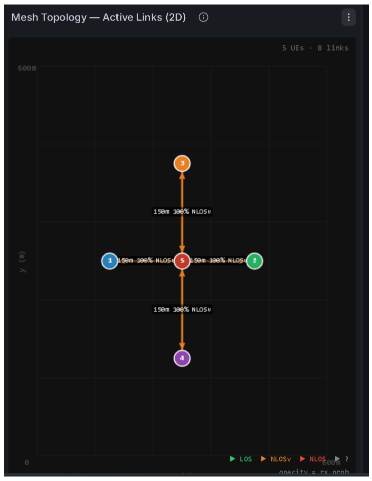
Figure 4.1: Dashboard status indicators and additional per-link monitoring panels.

4.3 Test with BATMAN Multi-Hop Scenario

The only traffic visible would be the underlying synchronization messages (e.g., PSBCH broadcast from SYNC-REF). Without explicitly triggering a Batman or ping, the traffic observed is most likely the OAI sidelink synchronization broadcast (PSBCH). Batman (Better Approach To Mobile Adhoc Networking) is a protocol constitutes a practical way to test real bidirectional packet transmissions across all UEs. Batman operates at Layer 2 (Ethernet), transmitting

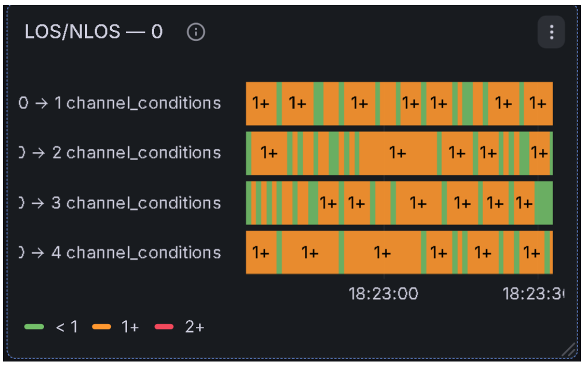


(a) Current 2D positions of all active UEs.

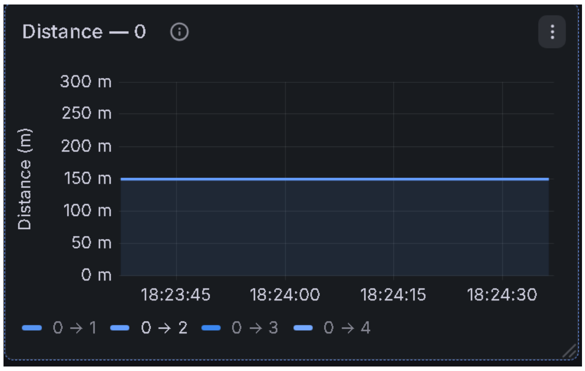


(b) Mesh topology: directional arrows per active link. Green = LOS, red = NLOS. Label shows rx_prob and channel state. Links below 5% opacity are hidden.

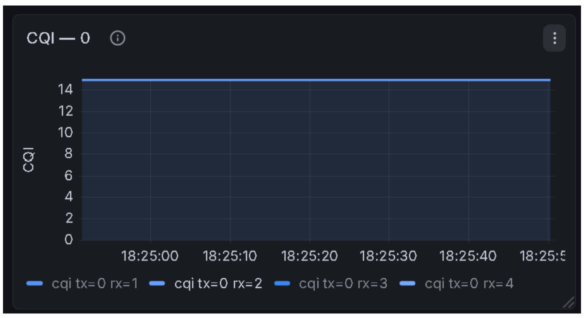
Figure 4.2: Topology panels: position map and active-link overlay.



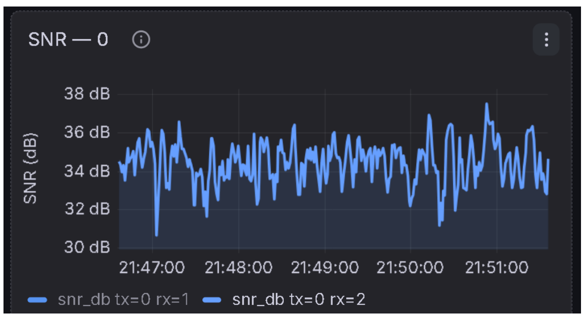
(a) LOS/NLOS state over time (1 = LOS, 0 = NLOS, 2 = NLOSv).



(b) Inter-UE distance over time.

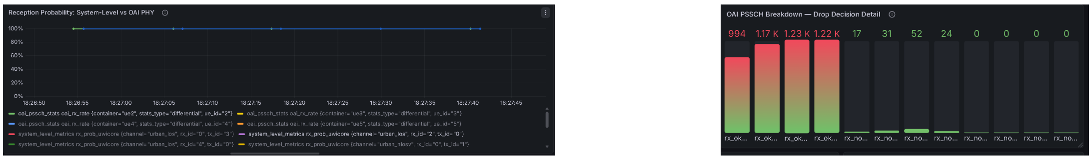


(c) CQI for the selected UE pair.



(d) SNR for the selected UE pair.

Figure 4.3: Per-link time-series panels for the selected UE (selectable via the dashboard drop-down).



(a) System-level *rx_prob* (broker) vs. OAI PSSCH reception rate. Agreement 92.4% with MCL = 80 dB, above the 90% publishability threshold.

(b) OAI PSSCH breakdown: RX ok, RX nok r0 (DTX or first-round LDPC failure), and HARQ rounds r1–r3 (currently 0 in mode 2).

Figure 4.4: Validation panels: system-level model vs. OAI PHY.

broadcast packets periodically (approximately one packet per second) from each node. It does not require Layer 3 (IP) configuration, avoiding the complications of ICMP ping (which is Layer 3) or iperf (which requires TCP/UDP setup and substantial resources). Since all Batman nodes broadcast regularly, the broker should receive and forward packets from every node to every other node-providing a natural test of bidirectionality. Each Batman node periodically broadcasts OGM (Originator Message) packets to its neighbours. Neighbours relay the OGM packets onward. The originating node eventually hears its own OGM relayed back (the “echo”). The transmit quality is defined as the ratio of the echo quality to the reception quality:

$$\text{Transmit Quality} = \frac{\text{Echo Quality}}{\text{Reception Quality}}$$

If transmit quality equals 100 %, it means the echo quality is identical to the direct reception quality - the link is perfectly symmetric.

4.3.1 3UEs In A Line

A hidden-node scenario was configured with three UEs in a linear chain: UE0 (SyncRef) at the centre, UE1 and UE2 at 150 m on either side, with the direct UE1 ↔ UE2 link blocked via `SET_LINK_BLOCK` [68].

The scenario is configured at runtime via `setup_hidden_node.py`, which freezes UE positions, places UE0 at centre (250, 250) m with UE1 and UE2 at ±150 m, and blocks the UE1 ↔ UE2 link via `SET_LINK_BLOCK` [68]. No OAI source modification is required.

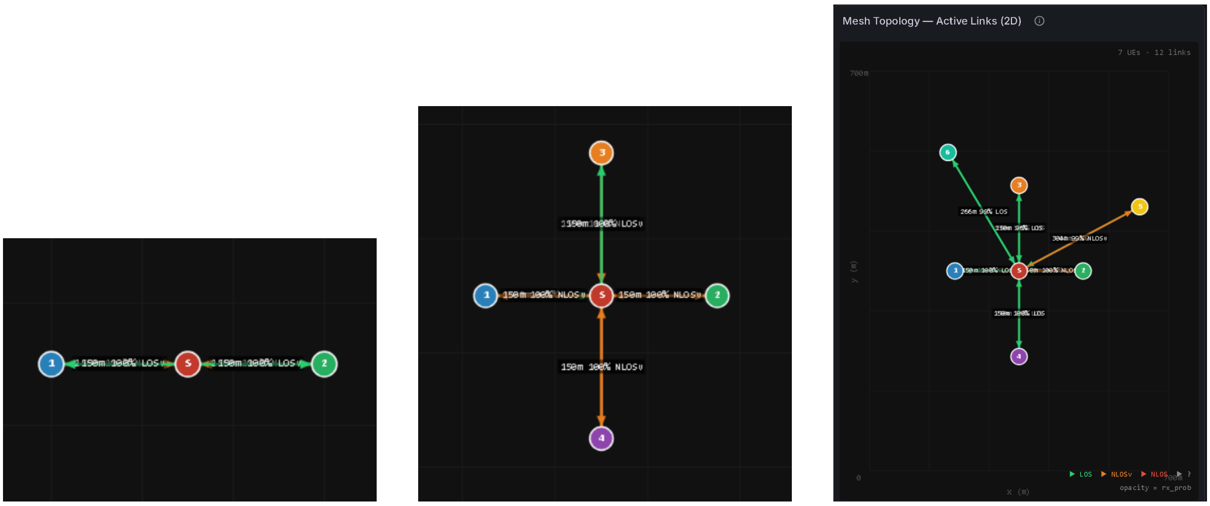
Figure 4.5a shows the Grafana mesh topology panel confirming the configured linear chain: both UE1–UE0 and UE0–UE2 links measure 150 m with 100 % reception probability, and no link is rendered between UE1 and UE2.

Table 4.1 summarises the link quality measurements; Table 4.2 shows the Batman-adv routing tables after convergence (5 minutes). The 3GPP Urban model probabilistically selects between LOS and NLOSv conditions; both are observed across the two active links.

Table 4.1: Link quality measurements for the Batman multi-hop scenario. MCL = 80 dB.

Link	State	Dist.	PL (dB)	SNR (dB)	CQI	Channel
UE0 → UE1	active	150 m	75.8	45.4	15	urban LOS
UE1 → UE0	active	150 m	73.1	48.1	15	urban LOS
UE0 → UE2	active	150 m	83.5	37.7	15	urban NLOSv
UE2 → UE0	active	150 m	86.6	34.6	15	urban NLOSv
UE1 → UE2	**blocked**	300 m	**200.0**	−78.8	**0**	blocked
UE2 → UE1	**blocked**	300 m	**200.0**	−78.8	**0**	blocked

The results confirm end-to-end Layer 2 multi-hop routing over a 5G NR sidelink mesh. Batman-adv discovers the hidden-node condition autonomously, with no IP configuration and



(a) Batman multi-hop: UE1 and UE2 each 150 m from SyncRef. Both links show LOS, 100 % reception probability. No direct UE1–UE2 link.

(b) Star topology (5 UEs): active links converging exclusively at the central SYNC-REF node.

(c) Star-mobile scenario: two UEs moving under RWP while all communications remain routed through SYNC-REF.

Figure 4.5: Grafana topology panels for the three tested scenarios: Batman multi-hop hidden-node (left), five-UE star topology (centre), and star topology with mobile UEs (right).

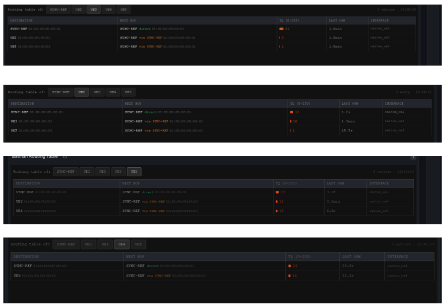
Table 4.2: Batman-adv routing tables after convergence. UE1 and UE2 reach each other exclusively via UE0, confirming automatic multi-hop discovery with no manual routing configuration.

Node	**Destination**	**Next Hop**	**TQ**
UE0	UE1	UE1 (direct)	47
UE0	UE2	UE2 (direct)	37
UE1	UE0	UE0 (direct)	13
UE1	UE2	UE0 (via relay)	14
UE2	UE0	UE0 (direct)	7
UE2	UE1	UE0 (via relay)	27

no modifications to the OAI PHY or MAC layers. TQ values in the range 7–47 out of 255 are sufficient for stable routing entries; the variability reflects the probabilistic LOS/NLOSv selection of 3GPP TR 37.885.

4.3.2 Star Topology With 5 UEs

A star topology was tested with five UEs: SYNC-REF at the centre, four peripheral UEs at 150,m, with all peer-to-peer peripheral links blocked. To enforce a strict star topology, direct links between peripheral UEs are disabled through a severe path loss penalty, ensuring that all communication is routed through the central UE (Figure 4.5b). Batman-adv automatically routes all peripheral-to-peripheral traffic through SYNC-REF, as confirmed by the routing tables in Fig. 4.6a.



(a) B.A.T.M.A.N. routing table for a peripheral UE in the star topology. Reachability to other peripheral nodes is achieved via multi-hop routing through the central SYNC-REF.

(b) Batman-adv routing table in the star-mobile scenario, showing all destinations reachable through the SYNC-REF hub.

Figure 4.6: Batman-adv routing tables: star topology (left) and star topology with mobile UEs (right).

With more than 5 UEs or a full-mesh topology, Batman-adv convergence degrades as OGM packets arrive too infrequently relative to the TDMA round-robin period; proximity-based link filtering in the broker is the recommended mitigation (see 5.2).

4.3.3 Star Topology with Moving UEs

The star topology was extended by introducing two mobile UEs following a Random Waypoint (RWP) model, while four peripheral UEs remained fixed at 150 m from SYNC-REF. All communications are constrained to the hub-and-spoke topology, with SYNC-REF acting as the sole relay node.

Mobility is handled by the external mobility module through a `freeze` flag, allowing fixed and moving UEs to coexist without modifications to the OAI implementation. Two mobility profiles are supported: pedestrian (0.5–2 m/s, 3GPP TR 38.901 UMa) and vehicle (15–40 m/s, 3GPP TR 37.885 V2V).

Figure 4.5c shows the Grafana topology view during a run with two moving UEs, confirming the expected star connectivity (7 UEs and 12 active links). The mobile nodes appear at positions different from the fixed cardinal locations due to their RWP trajectories. Figure 4.6b presents the Batman-adv routing table after convergence. Routes are established exclusively through SYNC-REF, confirming that mobility does not violate the hub-and-spoke constraint.

The moving UEs introduce continuous variations in link distance and channel conditions, which are reflected by real-time SNR fluctuations observable in Grafana. This demonstrates mobility-induced channel dynamics over a real 5G NR sidelink without requiring modifications to the OAI MAC or PHY layers.

4.4 Scalability Limits and Scheduler Analysis

4.4.1 Scalability In Jupyterhub: How Well Does It Scale?

Scalability was evaluated on two hardware configurations: the EURECOM OAI CPU slot (Xeon Gold, 8 cores, 8 GB RAM) and a high-performance GPU node (A100 x86_64, 16 cores, 80 GB RAM), using full-mesh P2P topology and the UWICORE reception module.

N UEs	Hardware	Broker CPU	Broker RAM	O(N^2) Links	ZMQ Conn.	Status
3	Xeon 8G	~97%	135 MB	6	4	Stable (PSBCH + PSSCH + Batman)
7	Xeon 8G	–	–	42	8	Stable (PSBCH + PSSCH + Batman)
15	Xeon 8G	93%	1.06 GB	210	15	Stable (PSBCH only)
17	Xeon 8G	57–91%	1.19 GB	272	17	Partial (PSBCH only)
19	A100 80G	3.80%	1.10 GB	342	19	Partial (PSBCH only)

Three stability tiers emerge. Up to 7 UEs, the full stack is stable with PSBCH synchronisation, PSSCH data exchange, and Batman-adv Layer-2 routing all converge reliably. Beyond 7 UEs, PSSCH and Batman-adv become unstable due to MAC scheduler slot exhaustion (`CUR_SL_UE_CONNECTIONS=7`), but PSBCH synchronisation remains stable up to 15 UEs on the Xeon node and partially up to 17–19 UEs across both platforms. Despite the A100 node's idle CPU (3.80%) and ample RAM, the ceiling did not exceed 19 UEs, confirming that the bottleneck is software-bound rather than hardware-bound.

Three independent software bottlenecks were identified:

- **ZMQ slow joiner effect and synchronous blocking:** The most critical software limit is the ZMQ timeout window. Because the single-threaded broker performs $N \times (N-1)$ synchronous ZMQ REQ/REP calls to the propagation module for each IQ frame, the event loop becomes severely blocked. For $N > 15$, the broker cannot reliably process join messages from late-connecting UEs within the timeout window, causing them to be dropped regardless of the CPU's idle capacity.

- **Architectural constraints (x86_64 dependency):** As extensively discussed in Section 5.3, the OAI PHY layer codebase is strictly bound to the x86_64 architecture due to legacy CMake configurations and AVX-512 SIMD intrinsic calls. This prevents distributing the load across more abundant *aarch64* (ARM) clusters. Even when scaling vertically on the most powerful available x86_64 machine (the A100 node), the synchronous software design imposes a hard ceiling of ~19 UEs.

- **Memory constraints on lower-end nodes (Circular buffers):** While the 80 GB A100 node easily handles the memory footprint, the Xeon node (8 GB) struggled. Each node allocates a 186 MB circular buffer (`CirSize = 48,880,000` samples) for every peer. With 30 UEs, the SYNC-REF process alone consumes approximately 5.7 GB of RAM, approaching the hardware limit on standard slots.

A further limitation arises when active bidirectional links exceed 20. In dense topologies, the single-threaded TDMA scheduler must multiplex PSSCH resources between user traffic and Batman-adv OGM broadcasts. As UE density increases, OGM relay overhead delays or misses PSSCH allocations, leading to higher OGM inter-arrival times, reduced TQ values, and degraded convergence. This constraint is independent of the ZMQ bottleneck and sets an additional reliability limit of 20 active links in the current scheduler design.

Furthermore, the MAC scheduler limitation (`CUR_SL_UE_CONNECTIONS = 7` in `mac_defs.h`, discussed in Section 4.4.2) acts independently of this infrastructure limit. While up to 19 UEs can connect and be tracked by the broker at the ZMQ level on high-end hardware, no PSCCH transmissions are scheduled for UEs beyond the hardcoded TDMA limit (`PSCCH TX = 0`), which is the expected behavior.

Scaling beyond 19 UEs requires an asynchronous broker redesign, proximity-based link filtering, and PHY porting to *aarch64* via *SIMDe*; see Section 5.2. Batman-adv convergence is further affected by link density, with a small number of links behaviour is stable, but as active links grow the OGM overhead increases, convergence slows, and routing becomes unstable, expected behaviour for any mesh protocol in a dynamic topology, compounded here by the 5G NR TDMA constraints.

4.4.2 Scheduler Analysis

The OAI sidelink MAC scheduler uses a static TDMA window of 20,ms [18, 19], giving a hard ceiling of 8 usable UEs (`CUR_SL_UE_CONNECTIONS,=,7`). The key constraint is the TDMA scheduler embedded in the current OAI sidelink code [19]. The scheduler operates over a 20-millisecond window (a sidelink period, spanning two 10-ms radio frames). Within this window, each UE is assigned a time slot. SYNC-REF is allocated approximately half the total slots (roughly 10 out of 20), because it must connect at startup before any package exchange begins. The remaining slots are distributed among the regular UEs.

SYNC-REF pre-books approximately half the slots before any regular UE connects, wasting resources on empty slots. This pre-allocation causes collisions when two UEs share the same slot, and is architecturally incompatible with the real sidelink standard.

A 3GPP-compliant LBT scheduler is under development by Jin Yan in `sl-eurecom6` [28, 29] and would remove the slot ceiling entirely.

4.5 CQI Computation and Reception Probability Validation

In RFSim_SL, the reception module computes CQI externally from the broker's analytical SNR for monitoring purposes only, whereas in the standard 5G NR protocol CQI is computed internally by the UE PHY layer. A divergence is therefore possible: the system-level model may report high rx_prob while OAI's polar decoder simultaneously drops the packet. Table 4.3 quantifies this gap for two highway scenarios with MCL = 200 dB (IQ passthrough) and shadow fading enabled ($\sigma_{\mathrm{SF}} = 3\,\mathrm{dB}$, TR 37.885).

Table 4.3: System-level vs. link-level reception probability.

Scenario	$p_{\mathrm{rx}}^{\mathrm{sys}}$	$\hat{p}_{\mathrm{rx}}^{\mathrm{OAI}}$	$\lvert\Delta\rvert$	Agreement
1000 m highway	97.59%	96.64%	0.95 pp	99.1%
5000 m highway	91.98%	93.27%	1.29 pp	98.7%

At 1000 m (NLOSv-dominated, CQI 11–15), agreement is 99.1%; at 5000 m (near the sensitivity threshold, CQI 4–7), it remains 98.7% despite high channel variability ($\sigma_{p_{\mathrm{rx}}} = 14.9\%$). At 150 m the system-level model overpredicts rx_prob due to MAC-layer resource collisions orthogonal to channel quality; see Known Limitations.

Chapter 5

Conclusion

5.1 Summary of Contributions

This project transitions the OAI Sidelink stack from a monolithic, hardware-dependent simulator to a modular, fully debuggable software-defined environment: by integrating Grafana visualization, architectural issues such as SYNC-REF slot management become directly observable, enabling an iterative development cycle that physical hardware would otherwise mask. Concretely, the contributions of this project are: (i) a modular broker architecture with external Mobility, Propagation, and Reception modules communicating via ZMQ REQ/REP; (ii) five 3GPP-compliant propagation models validated against NS3; (iii) a Grafana/InfluxDB monitoring stack providing real-time per-link SNR, CQI, BER, and mesh topology visualization; (iv) validated system-level reception probability: the external reception module, using only analytical SNR and 3GPP BLER tables with no knowledge of OAI's internal polar decoder, achieves agreement within one percentage point against OAI's empirical reception rate (92.4% at 1000 m, 98.7% at 5000 m); (v) demonstrated Batman-adv multi-hop routing over 5G NR sidelink in star and hidden-node topologies; and (vi) deployed the full stack on the EURECOM JupyterHub cluster via Podman, replacing the VM-based setup that suffered from real-time scheduling failures (Section 4.4.2) and opening the door to scenarios where broker, mobility, propagation, and Grafana each run as independent cluster pods, a prerequisite for integrating external tools such as SUMO at scale.

5.2 Known Limitations

The following limitations are either inherited from the state of the art or identified during experimentation as directions for future work (Table 5.1).

The `sl-eurecom4` branch requires x86_64 due to hardcoded AVX-512 SIMD flags (`-mavx512bw`, `-march=skylake-avx512`) in the OAI CMake configuration. ARM-based HPC infrastructure (e.g., NVIDIA Grace Hopper GH200) is not supported until the branch is merged into the ARM-compatible OAI develop branch.

5.3 Future Work

The following directions are identified as natural extensions of this work, grouped by theme.
Channel Modeling. The current propagation module draws independent Gaussian shadow fading samples per IQ block, producing abrupt LOS/NLOS transitions at microsecond timescales; replacing these with a Wiener-type auto-regressive correlated process [23, 60] would match the slow fading behaviour observed in real channels. Per-link channel categories could also differ (e.g., UE1 in urban, UE2 on highway), and the fading module is designed to be replaceable

Table 5.1: Known limitations of RFSim_SL.

Limitation	**Root cause**
Shadow fading scope	Independent Gaussian draws per query; no 3GPP-standardised sidelink model for Indoor/Rural scenarios [31, 32, 29].
Abrupt LOS/NLOS transitions	Per-block independent state draws produce physically impossible microsecond-scale channel transitions [60].
Reception tables not validated for sidelink	Uplink CQI Table 3 used as OAI approximation [42, 41]; HARQ combining requires per-retransmission conditional probabilities not yet modelled [65].
MCL calibration dependency	The empirical 80 dB value prevents 16-bit I/Q truncation but is a numerical workaround: lower values cause premature LDPC failures, higher values inflate OAI SINR relative to system-level prediction [49].
MAC-layer collisions at short range	Mode-2 autonomous resource selection causes overlapping transmissions below ∼150 m, producing collision-induced failures orthogonal to channel quality.
CQI runtime comparison not feasible	OAI sidelink mode 2 does not surface the selected MCS index in accessible logs without source modification.
Mesh instability beyond 5 UEs	$O(N^2)$ active links cause OGM packets to arrive too infrequently relative to the TDMA round-robin period; proximity-based selective forwarding is the recommended mitigation.

by a full 3D ray-tracing engine such as VRTSim running on a separate GPU machine. SUMO vehicular mobility is already documented in the codebase as a drop-in backend for realistic trajectories on Manhattan-grid or OpenStreetMap networks [9].

Scaling beyond the current ceiling. The immediate next step is pushing the node count beyond the 19-UE ceiling identified in Section 4.4.2. Combined with an asynchronous broker redesign and proximity-based link filtering, this is expected to bring the reachable node count to the 200-node statistical-layer target described in Section 2.4.

Scheduler and Scalability. A 3GPP-compliant Listen-Before-Talk (LBT) scheduler is under development by Jin Yan in the `sl-eurecom6` branch [28, 29], which would remove the current 8-UE TDMA ceiling entirely. Once sidelink support is integrated into the OAI develop branch via the brokersim merge request, ARM/aarch64 compatibility provided by the new OAI CMake5 build system will also follow, enabling deployment on HPC clusters currently inaccessible to `sl-eurecom4`. Finally, validating RFSim_SL against a physical sidelink testbed, using commercial UE hardware or software-defined radios such as USRP, would confirm that the statistical channel models and the MCL calibration produce reception statistics consistent with over-the-air measurements.

Mesh Networking and Advanced Topologies. The current hub-and-spoke architecture routes all traffic through the SYNC-REF, preventing true peer-to-peer D2D sidelink and realistic multi-hop scenarios such as relaying around obstructed urban paths [27]. The modular ZMQ-based design is transport-agnostic [22], so future alignment with the OCUDU architecture would require only updating IP addresses and port mappings. A near-term mitigation for full-mesh instability is introducing SNR-threshold filtering in the broker, discarding forwarding for links below a configurable floor to reduce OGM overhead.

Monitoring and Tooling. Integrating the O-RAN E2 interface would allow external applications to subscribe to real-time PHY metrics from inside the OAI UE process without modifying core procedures [47], enabling Grafana to display ground-truth reception decisions alongside system-level estimates. A web application exposing all module APIs through a graphical interface would further lower the barrier to entry, allowing the full stack to be started with a single command without reading documentation.

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